AuraCoach: An Integrated Ecosystem for Real-Time Biometric Telemetry and Interoceptive Biofeedback in Public Speaking

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Abstract

Public speaking anxiety frequently generates a "Metacognitive Blind Spot" where speakers lose the ability to objectively monitor their physiological state and speaking pace [1]. AuraCoach addresses this phenomenon by transforming invisible biometric data into actionable feedback via an integrated iOS, watchOS, and macOS ecosystem. This report documents the system's development lifecycle, from the Human-Computer Interaction (HCI) research—based on needfinding and prototyping—to its technical implementation. The architecture utilizes real-time heart rate and gesticulation monitoring to provide discrete haptic biofeedback. A core feature of the system is the "Heart Sync" intervention, which leverages interoceptive anchoring through slow-rhythmic tactile stimulation [2] to facilitate physiological self-regulation [4].

1 Introduction

Public speaking is widely recognized as one of the most prevalent triggers for acute anxiety. During high-stakes presentations, speakers frequently experience intense physiological arousal, manifesting as an elevated heart rate, erratic gesticulation, and an uncontrollably accelerated speech pace.

The primary challenge, however, is not the anxiety itself, but the speaker's diminished capacity to perceive it in real-time [10]. Eysenck et al. (2007) posit that anxiety impairs the efficient functioning of the goal-directed attentional system, significantly decreasing attentional control. This cognitive overload creates a "Metacognitive Blind Spot": the speaker's attention becomes heavily influenced by the stimulus-driven attentional system, causing a loss of awareness regarding their own physical and vocal pacing. Visual feedback systems, such as timers or secondary screens, often fail to mitigate this issue as they require the speaker to divert their gaze from the audience, further taxing a compromised working memory.

AuraCoach was designed to address this blind spot through the principles of Calm Technology and Affective Computing [8]. By utilizing an Apple Watch to discretely

monitor biometrics, an iOS application to orchestrate data, and a macOS dashboard for AI-driven behavioral analysis, AuraCoach acts as an invisible co-pilot. The system provides interoceptive haptic feedback [3]—bridging the gap between physiological distress and conscious self-regulation—allowing the speaker to regain control without breaking the connection with the audience.

2 Human-Computer Interaction (HCI) Process

The development of AuraCoach followed a rigorous User-Centered Design (UCD) methodology to ensure the system effectively addresses the real-world needs and cognitive limitations of public speakers. Our HCI workflow was structured into sequential, data-driven phases:

1. **Needfinding:** We conducted qualitative interviews with 10 individuals to explore their specific pain points, emotional triggers, and needs during public speaking. To validate and expand upon these insights, we subsequently distributed a survey that collected 18 quantitative responses.

2. **User Modeling:** The synthesized research data was used to define specific user archetypes (Personas), which guided our feature prioritization.

3. **Prototyping & Testing:** We developed prototypes using Figma, which were then tested with 5 representative users. Given the haptic-centric nature of AuraCoach—which is impossible to simulate fully on a screen—the critical feedback gathered from these sessions was applied directly to the final SwiftUI implementation.

2.1 Needfinding: Interviews and Surveys

To understand the specific pain points and contextual constraints of our target users, we conducted a mixed-methods needfinding phase. This consisted of 10 in-depth qualitative interviews followed by a quantitative survey distributed to 18 participants, encompassing university students, corporate employees, and recent graduates.

Qualitative Insights (Interviews): The interviews confirmed the hypothesis of the "Metacognitive Blind Spot." Participants consistently reported experiencing physiological symptoms such as tachycardia and shallow breathing, which translated into accelerated speech rates and erratic gesticulation. Crucially, when asked about their awareness during the speech, respondents admitted to having "zero awareness" or experiencing "blackouts," only realizing their altered pacing after the performance. Furthermore, participants expressed a strong desire for a post-event review mechanism, stating that analyzing objective data on a larger screen (e.g., a dashboard) would be significantly more beneficial than relying on vague peer feedback.

Quantitative Validation (Survey): The survey corroborated our qualitative findings and provided critical constraints for our system design:

- **High Stress and Low Awareness:** The majority of respondents rated their anticipatory anxiety at a 4 or 5 out of 5, while simultaneously rating their real-time awareness of speaking pace at remarkably low levels (1 or 2 out of 5).

- **The Constraint of Invisibility:** When asked about their biggest concern regarding the use of a digital coaching tool, the overwhelming majority cited "The audience noticing I am using an aid."
- **Feedback Preference:** Participants unanimously indicated that a "discreet physical sensation (e.g., a silent vibration)" would be the least distracting and most acceptable form of real-time intervention.

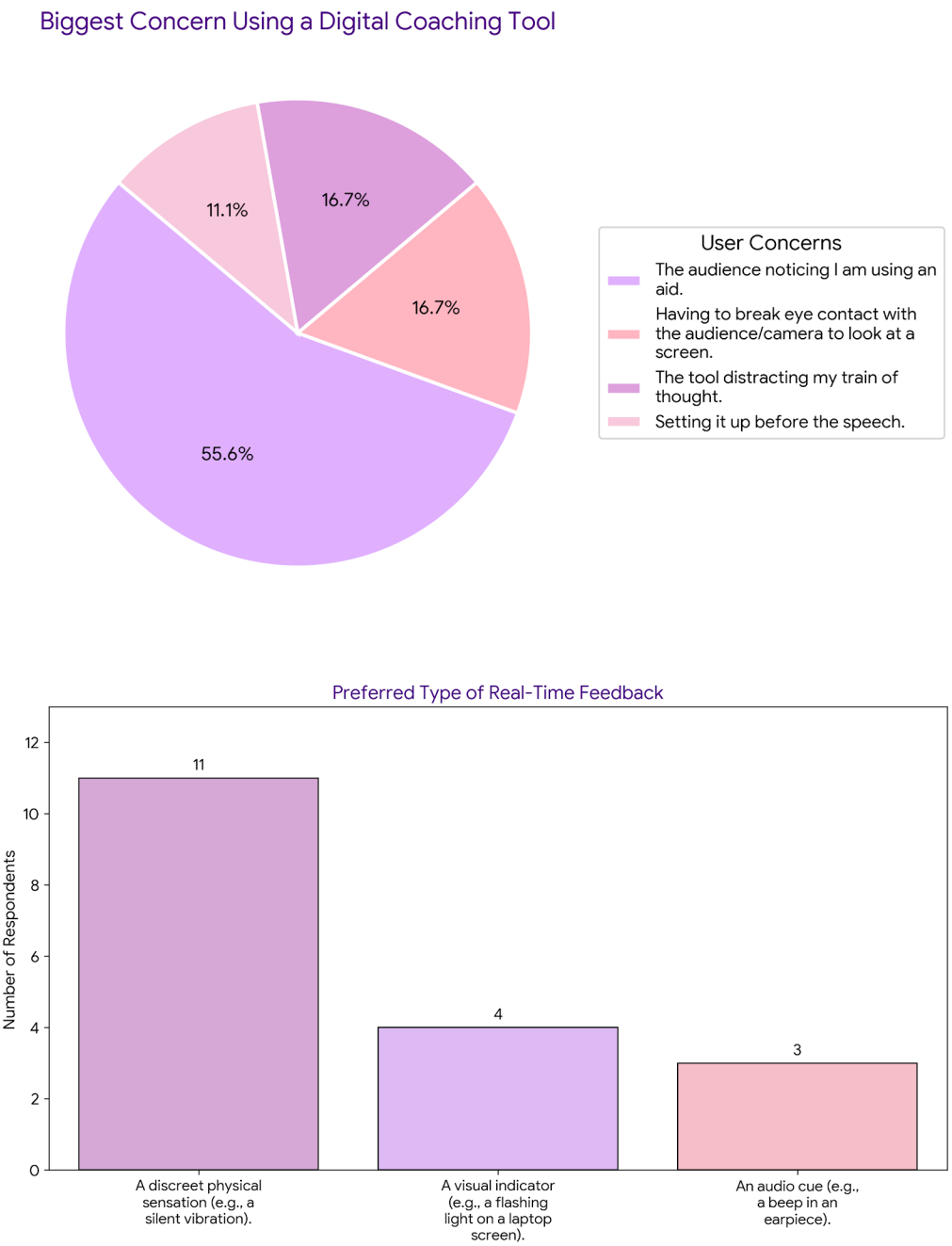

Figure 1: Survey results highlighting user concerns (top) and preferred real-time feedback modalities (bottom).

These findings established the core design requirements for AuraCoach: the real-time intervention must be strictly non-visual and haptic-based to preserve audience engage-

ment, while the complex data analysis must be deferred to an asynchronous, post-session desktop dashboard.

2.2 User Analysis: User Archetypes

Based on the synthesized needfinding data, we distilled our target audience into two primary user archetypes. This approach allowed us to address different facets of public speaking anxiety and define the specific requirements for the iOS, watchOS, and macOS modules.

Archetype 1: The "Blind-Spot" Sufferer Representing students and early-career professionals, this profile frequently engages in remote video interviews or academic presentations. Their primary pain point is the metacognitive blind spot: under stress, their heart rate spikes and they accelerate their speech to the point of losing their breath, yet they remain entirely unaware of this pacing until the session ends. They explicitly reject visual alerts, as shifting focus from the camera or audience breaks their concentration.
Design implication: Driven by this archetype, we prioritized the watchOS silent haptic feedback, allowing the user to receive an invisible pacing warning without breaking eye contact.

Archetype 2: The Executive Speaker Representing senior professionals, this archetype operates in high-stakes, live environments, such as boardroom meetings and executive pitches. In these formal contexts, interacting with a smartwatch or checking a smartphone screen is considered highly unprofessional. Their main concerns revolve around managing physical presence (e.g., erratic gesticulation) and acquiring objective data to review their performance with a clear head post-event.
Design implication: This profile shaped the development of the macOS desktop dashboard for asynchronous, AI-driven review, as well as the watchOS "Tactile Grounding" feature, which provides a discrete physical anchor to manage kinetic energy silently.

2.3 Prototyping and Evaluation

Following the definition of our user archetypes, we developed medium-fidelity interactive prototypes. The evaluation of these wireframes by our testing cohort highlighted crucial usability bottlenecks, which fundamentally reshaped our final implementation:

- **Cognitive Load in Session Initiation:** Testers expressed confusion over a "Double Start" mechanism, unsure whether to initiate tracking from the iPhone or the Apple Watch.
  Solution: We centralized the control architecture. The iPhone now acts as the primary "Local Brain," where pressing "Start Session" sends a silent background command to automatically wake the Watch sensors. The Watch interface was simplified to act purely as a passive status indicator, drastically reducing cognitive friction under stress.

- **Post-Session Data Depth:** While the mobile text report was appreciated, users found it lacked the statistical depth necessary to dissect their performance comprehensively.

Solution: This limitation drove the expansion of the ecosystem into a desktop macOS Dashboard. While the iPhone retains a quick summary, the desktop interface introduces deep data visualization (e.g., continuous line charts), allowing users to pinpoint the exact minute a panic spike occurred during their speech.

- **Contextualization of Raw Biometrics:** Testers noted that reading raw numbers (e.g., "123 BPM" or "4.6 G") lacked meaning without historical context, making it difficult to assess if a peak was an anomaly.
  Solution: We introduced dynamic contextualization through semantic coloring, shifting the UI from green to red based on severity to allow for pre-attentive processing. Additionally, the dashboard now actively compares current session peaks against the user's historical baseline.

- **Long-Term Engagement and Retention:** Evaluators pointed out a lack of intrinsic stimuli to maintain motivation after reviewing just a few sessions.
  Solution: To foster long-term practice, we integrated a Gamification layer. By introducing an "Achievements" module, the system shifts the psychological framing of public speaking from a stressful test to an interactive challenge, rewarding users for milestones such as maintaining low average anxiety or controlling speech pace.

3 System Architecture

3.1 The AuraCoach Ecosystem

To effectively address the metacognitive blind spot without increasing the speaker's cognitive load, the AuraCoach architecture is distributed across three hardware platforms, unified by a cloud backend. This distributed approach ensures that each device performs the task it is best suited for, minimizing latency and maximizing user focus. The ecosystem comprises four main pillars:

- **watchOS Application (The Sensory Hub):** Acting as the primary data collection and intervention point. It continuously tracks heart rate (via `HealthKit` [6]) and kinematic data (via `CoreMotion` [7]), while delivering discrete haptic biofeedback directly to the user's wrist.

- **iOS Application (The Orchestrator):** Serving as the local computational brain during a live session. It manages the session lifecycle, performs real-time speech analysis (Words Per Minute), and calculates a dynamic Anxiety Indicator by cross-referencing biometric streams.

- **macOS Application (The Analytical Dashboard):** Designed for asynchronous, low-cognitive-load review. It provides deep data visualization and interfaces with external AI models to generate actionable behavioral reports post-session.

- **Firebase Integration (The Cloud Backbone):** Acting as the central repository, it ensures secure, real-time cross-device data synchronization, allowing a session recorded on the iPhone/Watch to be immediately available for review on the Mac.

3.2 Data Management and Synchronization

The data flow within the AuraCoach ecosystem is divided into two distinct pipelines to balance real-time responsiveness with persistent storage:

1. Local Real-Time Telemetry (`WatchConnectivity`): During an active session, latency is critical. We implemented Apple's `WatchConnectivity` framework to establish a high-frequency, local communication channel between the Apple Watch and the iPhone. This allows the watch to stream raw BPM and G-force data to the phone, and conversely, allows the phone to trigger haptic interventions on the watch almost instantaneously, without relying on an internet connection.

2. Persistent Cloud Synchronization (Firebase Firestore): Upon the completion of a session, the localized data must be securely stored and formatted for deeper analysis. The iOS Orchestrator compiles the session metrics (including WPM, peak BPM, and average anxiety) and pushes a structured document to Firebase Firestore. The macOS dashboard continuously listens to this Firestore database, instantly populating the user's historical timeline as soon as a new session is finalized on the mobile device.

4 Module Implementation

4.1 iOS Application: Orchestration and Voice Analysis

The iOS application serves as the "Orchestrator" of the AuraCoach ecosystem, managing the session lifecycle and synchronizing data streams from both the user's speech and the wearable sensors.

Real-Time Speech Analysis (WPM): The core of the linguistic analysis is handled by the `SpeechManager` class, which utilizes the *Apple Speech Framework* to perform on-device transcription. To solve the challenge of monitoring speaking pace without visual distraction, we implemented a sliding-window algorithm that calculates Words Per Minute (WPM) in real-time. By tracking the count of transcribed words over the session duration, the system identifies "speech spikes" — moments where the speaker accelerates beyond their natural baseline, a common physiological response to increased cognitive load [1].

Multi-Modal Anxiety Algorithm: To quantify the user's stress level, we developed a weighted algorithm that aggregates different physiological and behavioral signals. The *Anxiety Score* (A) is calculated as follows:

$$A = (w_1 \cdot \Delta BPM_{norm}) + (w_2 \cdot G_{peak}) + (w_3 \cdot \Delta WPM_{norm}) \tag{1}$$

Where:

- ΔBPM_{norm} represents the heart rate deviation from the user's resting baseline.
- G_{peak} is the magnitude of physical agitation (tremors/fidgeting) captured by the accelerometer.

- ΔWPM_{norm} is the speech acceleration factor, calculated only when voice analysis is active.
- $w_{1,2,3}$ are dynamic weights adjusted to prioritize biometric stability over linguistic speed.

Modular Orchestration and User Agency: A key feature of the iOS Orchestrator is its modularity. Recognizing that different speaking contexts (e.g., a formal stage vs. a remote interview) require different levels of intervention, the app allows the user to toggle specific modules:

- **Voice Analysis Toggle:** The user can disable the `SpeechManager` (SFSpeech) to save battery or ensure privacy, in which case the anxiety algorithm re-adjusts its weights to rely solely on biometrics.
- **Haptic Feedback Remote Control:** Directly from the iPhone interface, the user can enable or disable the haptic pulses on the Watch. This allows the speaker to decide, mid-session, if they require active physiological entrainment or if they prefer a "silent tracking" mode without tactile prompts.

Session Management and AI Integration: Leveraging the `StorageManager`, the application compiles all telemetry into a structured JSON document stored in *Firebase Firestore*. Upon session completion, the `AIManager` handles the asynchronous request to the Groq API, passing the biometric timeline to the *LLaMA-3.3-70b* model. Rather than providing a generic summary, the AI acts as a digital speech coach, generating a structured report focused on four analytical dimensions:

- **Performance Overview:** A high-level assessment of the speaker's physiological and vocal control.
- **Biometric Patterns:** An analysis correlating heart rate spikes with physical agitation (G-force), identifying specific moments of somatic distress.
- **Speech Pacing Analysis:** An evaluation of WPM fluctuations to determine if the speaker maintained a consistent rhythm or succumbed to "anxiety-driven acceleration."
- **Actionable Advice:** Two to three concrete, data-driven tips for future improvement (e.g., specific breathing cues for identified panic spikes).

This ensures that the speaker receives an immediate, structured debrief before transitioning to the longitudinal analysis on the macOS dashboard.

Immediate Post-Session Summary: Once the session is finalized, the iOS application transitions from a recording interface to an immediate results dashboard. This view is designed to provide the speaker with a rapid "debrief" while the experience is still fresh. The summary includes:

- **Performance Highlights:** Clear visualization of maximum and average values for heart rate (BPM) and speaking pace (WPM), providing instant objective feedback on the session's peaks.

- **Temporal Anxiety Chart:** A simplified version of the anxiety timeline, allowing the user to visually identify the specific segments of the speech where stress levels were highest.
- **On-the-Fly AI Insights:** The natural-language report generated by *LLaMA-3.3* is displayed. This provides the user with immediate qualitative advice, bridging the gap between the live performance and the deeper, more strategic analysis that will take place later on the macOS dashboard.

4.2 watchOS Application: Biometrics and Haptics

The watchOS module is the primary interface for "Calm Technology" interaction. Unlike the other components of the ecosystem, it is designed for peripheral awareness [9], ensuring that the speaker remains anchored to their physiological state without visual distraction.

Biometric Sensing and Background Persistence: To ensure continuous data collection even when the user's wrist is lowered, we implemented a `HKWorkoutSession` through *HealthKit*. This architectural choice prevents the system from being suspended by the operating system, allowing `MotionManager` to stream high-frequency accelerometer data via *CoreMotion*. The magnitude of acceleration ($G = \sqrt{x^2 + y^2 + z^2}$) is calculated locally to detect tremors or erratic gesticulation, which are then transmitted to the iOS Orchestrator for anxiety scoring.

Tactile Grounding via Digital Crown: An interaction implemented in the watchOS app is the use of the Digital Crown as a "tactile fidget" tool. By mapping the crown's rotation to the `WKHapticType.click` API—in alignment with Apple's interface guidelines [5]—we provide the user with a mechanical-like feedback. This serves as a "grounding" mechanism: a discrete channel for discharging kinetic energy and nervous tension during a speech, completely invisible to the audience.

Interoceptive Reset: The "Heart Sync" Feature: Building on the research of *Azevedo et al. (2017)* regarding the calming effects of heartbeat-like tactile stimulation, we developed the "Heart Sync" (or Calm Mode). By performing a 1.5-second `LongPressGesture` on the watch face, the user activates a 60-second "Interoceptive Reset." During this window:

1. All stress-related alerts are muted to prevent further cognitive spikes.
2. The `WatchHapticManager` triggers a specific rhythmic sequence: three deep, double-pulse haptics (`.notification` type) at a fixed 60 BPM rate.

This intervention utilizes *physiological entrainment*: providing a slow, steady external "pulse" encourages the speaker's autonomic nervous system to synchronize with the wearable's rhythm, effectively lowering the actual heart rate through a subconscious feedback loop.

4.3 macOS Dashboard: Data Sense-Making and Multi-Level Analytics

The macOS application is the analytical hub of the ecosystem, designed to transition the user from real-time stress management to long-term behavioral reflection. While the mobile and wearable modules focus on "calm intervention," the macOS module is dedicated to "behavioral insight."

Global Dashboard: Longitudinal Progress and Retention: The primary interface provides a high-level overview of the user's evolution, which would be impossible to manage on a mobile device. By aggregating data from every session stored in *Firebase*, the system calculates:

- **Historical Baselining:** The dashboard establishes the user's "normal" speaking pace and average heart rate over time. This allows for the dynamic calculation of variances (e.g., "+15% BPM vs your average"), which is essential to help the user understand if they are structurally improving their stress management across multiple events.
- **Gamified Engagement:** To foster long-term practice, the system tracks *Behavioral Milestones*. By unlocking achievements based on biometric stability, the user is motivated to view public speaking not as a repetitive source of dread, but as a measurable "skill-tree" to be mastered through consistent training.

Session Deep-Dive and Behavioral Legacy: When selecting a specific past session, the interface provides a granular post-session view. This level of analysis is crucial for overcoming the metacognitive blind spot:

- **Integrated post-session analysis:** The dashboard provides a granular review by mirroring the performance highlights of the iOS post-session summary. This is integrated with dedicated *Swift Charts* for the Anxiety Score (visualized via a chromatic heatmap) and Heart Rate (BPM), alongside a *Key Moments* module for peaks identification.
- **Asynchronous AI Mentorship:** The AI-generated report created at the end of each session is permanently stored and accessible on the Mac. This allows the user to re-read specific behavioral suggestions (generated by *LLaMA-3.3*) even weeks after the event. Having the report side-by-side with the data charts transforms the AI's advice into a concrete educational tool, allowing for an analysis of the triggers when the speaker is no longer under emotional pressure.

System Rationale: The architectural separation between the "recording phase" and the "analysis phase" (macOS) is a deliberate design choice. It ensures that the speaker is never tempted to engage with complex analytics while on stage—preserving the principles of *Calm Technology*—while providing a professional, dedicated space for data-driven self-correction once the performance is over.

5 Conclusion

The development of AuraCoach demonstrates how an integrated, multi-platform ecosystem can effectively mitigate the "Metacognitive Blind Spot" inherent in high-stress public speaking. By grounding our design choices in established psychological frameworks—specifically *Attentional Control Theory* and *Rhythmic Entrainment*—we shifted the paradigm of speech coaching from traditional, distracting visual aids to discrete, interoceptive biofeedback.

Throughout the HCI process, we successfully navigated the tension between data density and cognitive load. The resulting architecture elegantly separates the "performance phase" (watchOS and iOS), which prioritizes calm, non-intrusive intervention, from the "reflection phase" (macOS), which leverages AI-driven analytics and longitudinal tracking for strategic self-improvement.

Final testing confirmed that the "Heart Sync" and "Tactile Grounding" features provide users with a sense of agency over their physiological responses, turning invisible anxiety into a manageable variable. AuraCoach proves that when wearable technology is designed through the lens of *Calm Technology*, it does not merely monitor the user; it actively mentors them, fostering a more conscious and confident presence in formal communication.

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